

Geothermal cable specifications

Flat power cables for submersible motors. Construction, dimensions and electrical properties.

Parameter	REPN-FX	REPN-QX
Cable designation	REPN-FX-5KV	REPN-QX-5KV
Rated voltage	5 kV	5 kV
Conductors	3 × 33.5 mm ² / 2 AWG	3 × 33.5 mm ² / 2 AWG
Nominal conductor diameter	6.54 mm	6.54 mm
Conductor material	Copper / TU1M07	Copper / TU1M07
Specified temperature rating	232 °C	204 °C
Insulation	Polyimide composite film + FEP	Polyimide composite film + EPDM
Individual core sheath	FEP	Lead
Cushion layer	PET fibre tape, 0.17 mm	PET fibre tape, 0.17 mm
Armour	Stainless steel tape	Stainless steel tape
Armour tape thickness × width	0.5 × 12.7 mm	0.5 × 12.7 mm
Maximum overall size (thickness × width)	14.1 × 35.7 mm	17.4 × 44.5 mm
Maximum conductor DC resistance at 20 °C	0.54 Ω/km	0.54 Ω/km
Minimum insulation resistance at 15.6 °C	5,000 MΩ·km	1,290 MΩ·km

Confirm allowable current, fluid compatibility and installation clearance.

Temperature values are specified cable ratings. Select the construction for the complete system and well conditions.

Oil & Gas cable specifications

5 kV AC / 50 Hz reference construction for oilfield submersible motors.

Construction & temperature

Parameter	Reference construction
Conductors	Three solid copper conductors
Insulation	Ethylene propylene rubber (EPR), nominal thickness 2.0 mm
Core sheath	Lead around each insulated core, nominal thickness 0.85 mm
Wrapping	Non-woven tape
Armour options	Galvanized steel tape; corrosion-resistant steel or Monel alloy
Continuous conductor temperature	Maximum 230 °C

Dimensions

Cores × area, mm ²	Nominal conductor Ø, mm	Maximum thickness × width, mm
3 × 10	3.56	12.8 × 33.1
3 × 13.3	4.11	13.5 × 35.1
3 × 16	4.50	13.8 × 36.2
3 × 21.15	5.19	14.5 × 38.3
3 × 25	5.65	14.9 × 39.5
3 × 33.6	6.55	15.8 × 42.4
3 × 35	6.60	15.9 × 42.5

These dimensions belong to the 5 kV reference construction. Confirm the supplied configuration, motor connection and installation clearance.

Oil & Gas operating environment

Fluid compatibility for the 5 kV reference construction.

Parameter	Galvanized-steel armour	Corrosion-resistant armour
Production-water pH	5.0–8.5	3.0–9.0
Maximum hydrogen sulphide	0.01 g/L	1.25 g/L
Maximum hydrostatic pressure	40 MPa	40 MPa
Maximum water content	100%	100%

Additional fluid limits

The following maximum concentrations apply to the corrosion-resistant reference construction.

Constituent	Maximum concentration, g/L
Carbon dioxide (CO ₂)	1.15
Chloride (Cl ⁻)	75
Bicarbonate (HCO ₃ ⁻)	1
Calcium (Ca ²⁺)	9
Sodium + potassium (Na ⁺ + K ⁺)	40

Match armour and cable construction to the well chemistry and operating conditions. The current rating depends on the surrounding medium and temperature; see the following page.

Oil & Gas current ratings

Maximum continuous current in A. Temperature values below refer to the surrounding medium.

Oil well fluid

Ambient °C	10 mm ²	16 mm ²	21.15 mm ²	25 mm ²	33.6 mm ²	35 mm ²
40	118	157	187	207	228	255
60	112	148	177	196	216	241
80	105	139	166	184	202	226
100	98	130	155	171	188	211
120	90	119	142	158	173	194
140	81	108	129	143	157	175
160	72	95	113	126	138	155
170	66	88	105	116	128	143
175	63	84	101	112	123	137
180	60	80	96	106	117	131
200	47	62	74	82	91	101
220	27	36	43	48	52	58
228	12	16	19	21	23	26

Gas-air environment

Ambient °C	10 mm ²	16 mm ²	21.15 mm ²	25 mm ²	33.6 mm ²	35 mm ²
40	96	127	151	167	182	204
60	91	120	143	158	173	193
80	86	113	134	148	162	181
100	80	105	125	138	151	169
120	73	97	115	127	139	155
140	66	88	104	115	126	140
160	59	77	92	101	111	124
170	54	72	85	94	103	115
175	52	68	81	90	98	110
180	49	65	77	86	94	105
200	38	51	60	66	73	81
220	22	29	35	38	42	47
228	10	13	15	17	19	21

230 °C is the conductor limit, not the ambient temperature at full load. Request a confirmed current rating for 13.3 mm². Final allowable current depends on the installation.